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# Global analysis of geological $CO_2$ storage by pressure-limited injection sites

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ABSTRACT

Limiting global warming to a 2 °C rise may require large-scale deployment of carbon capture and storage (CCS). Due to the key role CCS plays in integrated assessment models of climate change mitigation, it is important that fundamental physical constraints are accounted for. We produce a global estimate of $CO_2$ storage resource that accounts for pressure-limits within basin-scale reservoir systems. We use a dynamic physics model of reservoir pressurisation that is sufficiently simple to be incorporated into energy systems models. Our estimates address regionally inconsistent methodologies and the general lack of consideration for pressure limitations in global storage resource estimates. We estimate a maximum pressure-limited resource base and explore scenarios with different injection patterns, and scenarios where the extent of CCS deployment is limited by the history of regional hydrocarbon exploration and the readiness of countries for deployment. The maximum pressure-limited global storage achievable after thirty years of injection is 3640$GtCO_2$ (121$GtCO_2yr^{-1}$), increasing to 5630$GtCO_2$ (70 $GtCO_2yr^{-1}$) at the end of the century. These represent an update to volumetric-based estimates that suggest in excess of 10,000Gt of storage resource available. When CCS deployment is limited to the top ten countries ranked by the GCCSI Storage Readiness Index, our maximum storage estimate decreases to 780$GtCO_2$ (26$GtCO_2yr^{-1}$) at the mid-century and 1177$GtCO_2$ (15$GtCO_2yr^{-1}$) at the end of the century. These latter results fall within the range of projected deployment by the IPCC and IEA and suggest that reservoir pressurisation will limit CCS deployment if development does not rapidly expand beyond the current implementation.

## 1. Introduction

Techno-economic models that meet net zero targets project rapid rates of carbon capture and storage (CCS) deployment (IPCC, 2018). Quantitative estimates for the total required storage range from 4 $GtCO_2$ to 10 $GtCO_2$ removed annually by 2050 (IEA, 2013, 2020; GCCSI, 2020; IPCC, 2023). Even the lower end of this scale requires more than 100 times the current rate of CCS deployment within 25 years (Scott et al., 2013; Martin-Roberts et al., 2021). Carbon capture and storage is a driver for total mitigation costs in energy systems models. It is therefore important that deployment of the technology is modelled realistically, with fundamental geological constraints accounted for in model projections.

Numerous approaches have been used to project pathways for large-scale $CO_2$ storage. Currently CCS projects are located in hydrocarbon-producing basins (Herzog, 2017) which allows the ratio of annual $CO_2$ injection to volumetric flow data to be used as a proxy for the scaleup of CCS. This proxy highlights greater prospects for CCS in North America and Europe, where historically there is an established hydrocarbon industry, and suggests prospects in India and China are more challenging (Lane et al., 2021). Other measures of industry, such as rates of well construction, have also been used as a projection for the scaleup of $CO_2$ storage. The industry approach suggests that to meet targets, continental-margin settings will need to host between 10,000 and 14,000 wells injecting $CO_2$ (Ringrose and Meckel, 2019). Carbon storage deployment can also be constrained by economic models based on financial drawbacks and logistical complications. Grant et al. (2022) combined an economic model with a global energy system model and, like other proxies, found CCS deployment to be more challenging in China and India. Another approach is to constrain the technological scaleup of CCS technologies using technoeconomic models of growth, which are frequently used to assess natural resource consumption. This approach has been used globally by Zahasky and Krevor (2020), and regionally for Europe by Zhang et al. (2022) and the USA by Zhang et al. (2023). Technoeconomic models of growth suggest that current projections for scaleup to 2050 are unfeasibly rapid, but that carbon storage

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deployed over the course of the century could be commensurate with cumulative contributions projected in the IPCC reports. However, technoeconomic models of growth do not account for the physical constraints of reservoir systems.

Injectivity constraints and the associated pressure build-up in storage reservoirs has been identified as a limiting factor for rates of $CO_2$ storage both at the scales of individual fields, and across regional resource plays (Birkholzer et al., 2009; Mathias et al., 2009; Huang et al., 2014; Bandilla et al., 2017; Szulczewski et al., 2014). If injection patterns raise pore pressures until they exceed the fracture gradient, then projects risk inducing seismicity or deformation of the caprock. Felt seismicity is a major concern. Seismicity and deformation in the subsurface could also compromise seals and allow $CO_2$ to leak back to the atmosphere. The risk of leakage is a significant barrier to public acceptance of CCS technologies (Hsieh and Bredehoeft, 1981; Rutqvist, 2012; Zoback and Gorelick, 2012; Vilarrasa and Carrera, 2015). For these reasons injection permits set limits on the maximum pressure at which CCS injection projects can operate (NSTA, 2023).

Detailed numerical models constrained by pressure limits have been used to estimate storage resource for case study examples but are too computationally intensive to be involved in regional storage assessment (Bandilla et al., 2012; Edwards ed at., 2015). Less computationally intensive approximate solutions for pressure fields produced by $CO_2$ injection have been derived from fluid dynamics equations (Nordbotten et al., 2005; Mathias et al., 2015). These principles form the basis for simple dynamic models that provide insight into more widespread pressure limitations. Dynamic models identify pressure constraints, and not plume migration, as the dominant factor limiting injection rates during the initial phase of $CO_2$ injection (Szulczewski et al., 2012, 2014). In Szulczewski et al. (2012), they were applied to analysis of a dynamic pressure-limited $CO_2$ storage resource base for the USA. The simple, analytical simulation tool EASiTool was developed to produce a fast storage capacity estimate that incorporates pressure constraints into estimations of geological carbon storage resource in saline aquifers (Ganjdanesh and Hosseini, 2017, 2018). Analytical techniques typically used for oil and gas aquifers have been extended to analyse the storage resource of pressure-limited storage aquifers. The Havnsø saline aquifer has been used as a case study to demonstrate these analytical processes (Thibeau and Adler, 2023).

The dynamic, analytical tool called CO2BLOCK was developed to incorporate pressure constraints into $CO_2$ storage resource estimates and allow the evaluation of varying injection patterns (De Simone et al., 2019; De Simone and Krevor, 2021). CO2BLOCK uses the principles of superposition in a similar way to EASiTool, however CO2BLOCK also accounts for nonlinear pressure response with $CO_2$ injection, and has the option to make a correction for the errors associated with superposition. CO2BLOCK has been successfully demonstrated on selected UK offshore reservoirs (De Simone and Krevor, 2021) and in other regional studies (Karvounis and Blunt, 2021; Qin et al., 2023).

In this work we extend analyses of storage resource limited by injectivity from regional evaluations to a global estimate. We use the sedimentary basin as our unit of analysis, making use of extensive databases of properties from fields in these basins. Our storage resource estimates represent an improvement upon current estimates as they are dynamic, or time-dependent and our global analysis is internally self-consistent. This avoids the major limitations of current global compilations of storage resources that rely on country-based assessments that have been made using a variety of methods.

## 2. Data and methods

### 2.1. Estimating pressurisation in reservoir systems

There are a number of existing simplified physics models that could in principle be used to make large-scale storage resource estimates and could be integrated into energy systems models. The simplest models treat the reservoir system as a closed box with static pressure buildup estimates (Zhou et al., 2008). Dynamic models capture the evolution of the reservoir pressures of the storage resource. Multiple dynamic models have been produced to solve multiphase fluid flow equations for a single injecting well (Nordbotten et al., 2005; Mathias et al., 2009; Vilarrasa et al., 2010; Juanes et al., 2010; Golding et al., 2011). Well models involving multiple injecting wells have been developed by superimposing the solutions of simple single well models onto themselves. These multiple well models can be used to evaluate reservoir pressure scenarios (Huang et al., 2014; Joshi et al., 2016). De Simone et al. (2019) calculate the superposition of single well solutions proposed by Nordbotten et al. (2005) and determine the error due to superposition empirically. This approach was implemented in the open-source and publicly available analytical tool CO2BLOCK that is used in this study.

We use CO2BLOCK to model the pressure buildup due to simultaneous $CO_2$ injection at multiple sites. In addition to the pressure buildup from injection, the tool makes estimates for two possible limits on injection pressure: tensile failure and shear failure of the reservoir rock. (De Simone et al., 2019; De Simone and Krevor, 2021). Note that although the model is constructed based on a conceptual picture of an injection well, it is equivalent to the representation of pressurisation from multiple injecting wells in close proximity. Thus, each injection point in our use is representative of a storage site which may in practice deploy multiple injection wells within a few kilometres of each other (Duong et al., 2019; Trupp et al., 2021; Department for Business, Energy and Industrial Strategy, 2022). The output is an estimate of the maximum rate of injection that can be sustained over a given time period, subject to pressure limitations, and the corresponding cumulative $CO_2$ mass stored.

There are several key assumptions built into the model. The reservoir is assumed to be homogeneous with injection sites spaced uniformly on a Cartesian grid and all operating with the same injection rate, which is constant in time. While a notional $CO_2$ plume geometry is estimated as a part of the pressure build-up calculation, CO2BLOCK does not use this information to estimate limits due to plume migration. No estimates are made of the extent of structural, stratigraphic, residual, dissolution or mineral trapping. We assume a maximum injection rate per storage site of 5 $Mtyr^{-1}$ reflective of ongoing and planned industrial scale storage deployment.

For a given basin or reservoir, there are a minimum of seven input parameters needed to for an estimate of the maximum dynamic storage resource. These are: (1) minimum and (2) mean reservoir depth, (3) mean porosity, (4) mean permeability, (5) mean reservoir thickness, (6) reservoir or site area and (7) the time duration of injection. Boundary conditions of the geological units must be defined as open or closed. The tool is flexible and if additional data are known they may be incorporated into CO2BLOCK for more accurate estimates. Optional data include rock and water compressibility, $CO_2$ density and viscosity, water viscosity, pore pressure, mean pressure, mean temperature, brine salinity, maximum principal stress, the ratio of mean horizontal stress to vertical stress, friction coefficient, cohesion of reservoir rocks and tensile strength. If these data are not defined, CO2BLOCK will adopt default values or estimates. For this study, we incorporated pressure, temperature and salinity data where it was present in two commercial databases described in §2.2. We define all basin boundary conditions as closed.

The calculations carried out by CO2BLOCK are summarised in Fig. 1 and relevant parameters are defined in Table 1. There are three outputs: maximum sustainable pressure of a reservoir system (MPa), maximum sustainable injection rate per site (Mt $yr^{-1}$) for the time duration, and storage resource of a reservoir (Gt) for the time duration. These outcomes are produced for varying spacings between injection sites.

### 2.2. Geological data

Sedimentary basins are the smallest geological entity considered in our analysis. We use basin-averaged values as input geological

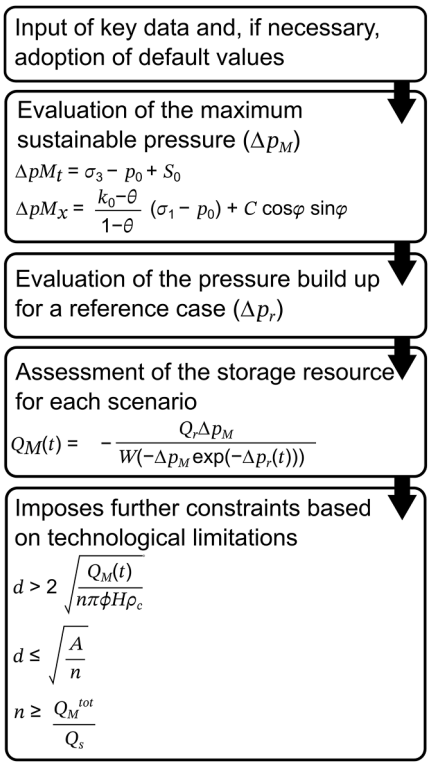



**Fig. 1.** Flow diagram summarising the steps taken by CO2BLOCK. Simplified from De Simone and Krevor (2021).

**Table 1**
Key parameters used in the CO2BLOCK calculations highlighted in Fig. 1.

| Parameters | |
|---|---|
| $\Delta pM_t$ | Pressure limit for tensile failure |
| $\sigma_3$ | Minimum principal stress |
| $p_0$ | Initial pressure |
| $S_0$ | Tensile strength |
| $\Delta pM_x$ | Pressure limit for shear failure |
| $k_0$ | Stress ratio |
| $\theta$ | Derived from friction angle |
| $\sigma_1$ | Maximum principal stress |
| $C$ | Cohesion factor |
| $\varphi$ | Friction angle |
| $\Delta p_r$ | Pressure response to a reference injection rate |
| $Q_M(t)$ | Injectivity |
| $Q_r$ | Reference injection rate |
| $W$ | Lambert function |
| $d$ | Distance between injection sites |
| $n$ | Number of injection sites |
| $\phi$ | Porosity |
| $H$ | Reservoir thickness |
| $\rho_c$ | $CO_2$ density |
| $A$ | Reservoir Area |
| $Q_M^{tot}$ | The total injection rate, including all sites |
| $Q_s$ | Injection rate limited by technical constraints |

parameters for CO2BLOCK. Thus, reservoirs at different stratigraphic levels in a basin are aggregated into a single unit. This is a basin-sized unit with the hydraulic properties that are the averages of the reservoirs within the basin. Any compartmentalisation within a basin by structures such as sealing faults or low permeability zones within reservoirs is ignored. This study aims to estimate storage resource by applying a consistent methodology to 248 basins globally. Accessing reservoir models that highlight heterogeneity and sealing fault architecture for this many basins is challenging. Therefore, we focus on a applying a simple model consistently, and consider more granular geological detail to lie beyond the scope of this study.

We adopt the basin scheme described by Evenick (2021), which outlines 764 sedimentary basins around the world based on multiple geodata sources including surface geological maps, basin-fill thickness maps, maps of basement structures, digital elevation models, gravity and magnetic surveys, and well data. This ensures storage resource estimates are defined at the highest level by geology rather than geopolitical boundaries. Regional maps of these basins are available in the Supplementary Data.

Geological data were obtained from over 80,000 petroleum reservoirs from the Wood Mackenzie field reservoirs database held within their Lens Subsurface solution. Due to the large number of entries, this database is useful for estimating geological parameters such as average porosities and permeabilities of potential storage units in a basin. In addition to field data, Wood Mackenzie provide their wells database. This allows us to analyse the distribution of wells and limit scenarios further by excluding basins with no history of petroleum exploration from resource estimates.

Open access geological data from the National Assessment of Geologic Carbon Dioxide Storage Resources provided inputs for basins in North America and central Africa (U.S. Geological Survey Geologic Carbon Dioxide Storage Resources Assessment Team, 2013; Warwick et al., 2013). To ensure the model was provided with essential inputs, in some regions a combination of data was used from the Wood Mackenzie database and C&C Reservoirs' Digital Analogue Knowledge System (DAKS) database (C&C Reservoirs, 2018). In the case of any discrepancies, the Wood Mackenzie data is prioritised to ensure results are as consistent as possible. A list of individual basins and their contributing data sources is supplied in the Supplementary Data.

The geological data sourced from these databases are filtered so that reservoirs are limited to those with a top depth greater than 800 m. This depth requirement ensures $CO_2$ remains in a supercritical state once stored in the subsurface (Benson et al., 2005). Geological data are then sorted by sedimentary basin and arithmetically averaged. For each basin, this produces the mean geological values such as mean porosity that are required inputs for CO2BLOCK. Results are presented per continent and per country. Where a basin straddles national boundaries, it is assigned to the country or continent that contains the majority of the basin area.

### 2.3. Defining scenarios and pressure-limited basins

To evaluate the impact different injection set ups have on our ability to meet CCS targets we utilise different injection 'scenarios'. We first created estimates of maximum storage for twelve combinations of numbers of sites per basin and site spacing, applied uniformly to each basin (Table 2). Scenario 3 mirrors the development plans for the Endurance project, with four clusters of injection wells spaced tens of kilometres apart (Dept. for Business, Energy and Industrial Strategy, 2022). For the upper bound spacing we follow the approach of Birkholzer and Zhou (2009) and apply a 30 km spacing between sites. This is a compromise between better interference spacing (Fig. 2) and the injection scenario being sufficiently small to fit within basins. The maximum of 400 injection sites is a similar number to the 542 oil and gas fields in the mature, well-developed Northern North Sea Basin. Scenarios with a greater number of sites and larger site spacing are more

**Table 2**
The number of injection sites and spacing between injection sites per basin that define each scenario.

| | | Distance between injection sites | | |
|---|---|---|---|---|
| | | 5 km | 15 km | 30 km |
| **Number of injection sites** | **4** | Scenario 1 | Scenario 2 | Scenario 3 |
| | **25** | Scenario 4 | Scenario 5 | Scenario 6 |
| | **100** | Scenario 7 | Scenario 8 | Scenario 9 |
| | **400** | Scenario 10 | Scenario 11 | Scenario 12 |

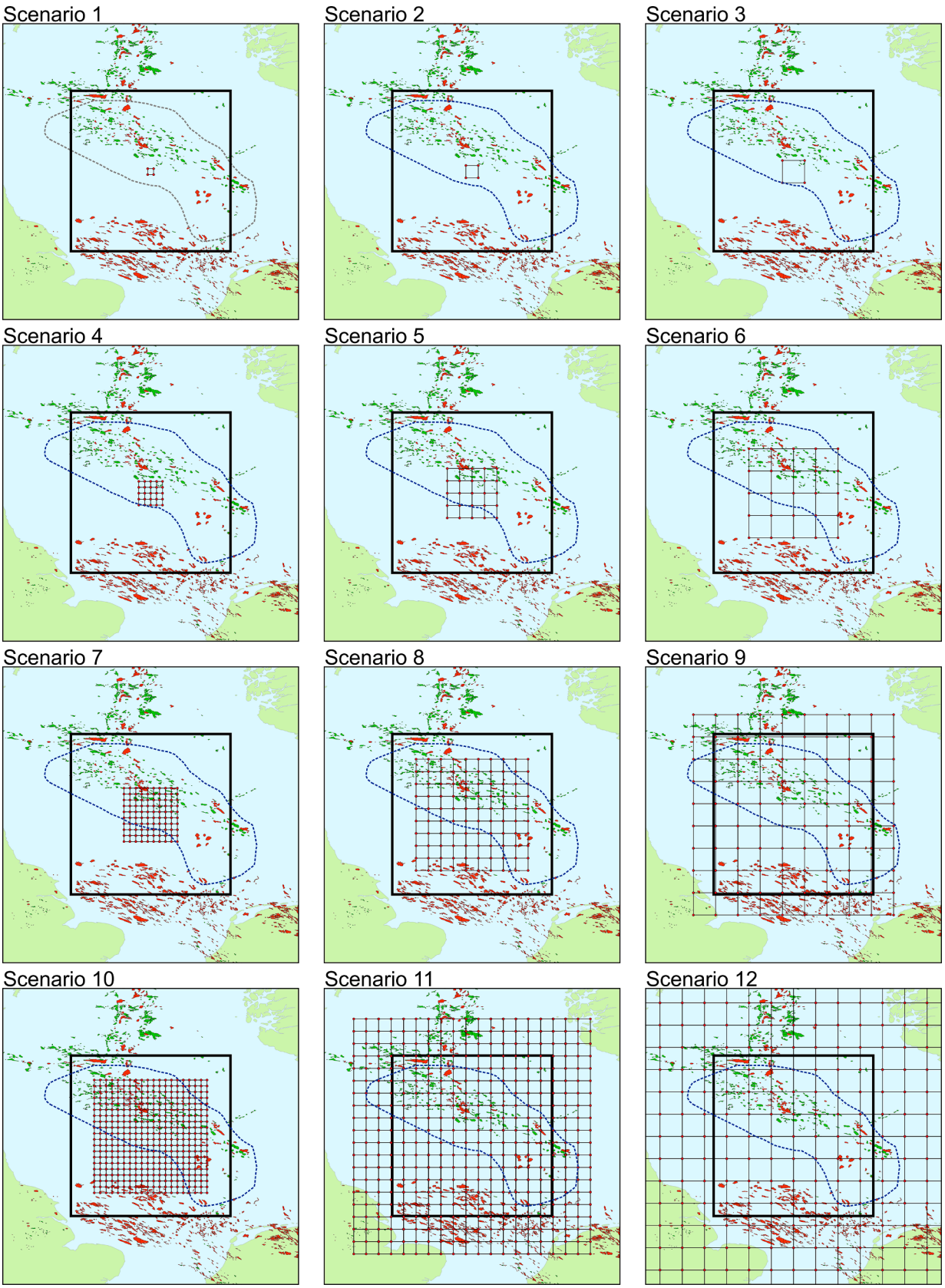



**Fig. 2.** The setup for each of the 12 scenarios when applied to the Central North Sea Basin. The basin itself is outlined by a dotted line, and the area of this polygon is equal to the area of the box used by CO2BLOCK. Injection sites are located at every node of the grids within the black boxes. Where the area of an injection scenario is greater than the area of a basin (e.g. Scenarios 9, 11 and 12), we explore an additional scenario (Scenario M). Source: Wood Mackenzie.

difficult to apply to smaller basins (Fig. 2). For example, Scenario 12 is only applicable for 20 basins, out of the 265 basins for which we have sufficient data for analysis.

From these estimates from uniformly applied injection site patterns, we construct a subset of scenarios of particular interest. Firstly, we construct a maximum scenario, Scenario M, in which the maximum storage resource estimate produced for each basin is used. Unlike scenarios 1 – 12, in Scenario M different basins employ different injection patterns. A breakdown of which scenarios of injection patterns are amalgamated in Scenario M is shown in Fig. 3.

Next, we construct scenarios based on socioeconomic factors that may limit the rate of CCS deployment. Areas with no history of hydrocarbon exploration have more uncertainty surrounding the geological character of the subsurface. This can lead to bottlenecks when developing new CCS projects. The Wood Mackenzie wells database highlights the location of known wells and we have counted these within each basin. We run a scenario where any basin with fewer than 100 hydrocarbon wells is excluded from our calculations to reflect more rapid CCS deployment in regions with historic hydrocarbon exploration. This reduces the number of eligible basins to 204 (Fig. 4).

Another barrier to CCS deployment is governmental support and the readiness of countries for deployment. The Global CCS Institute identify countries ready for imminent deployment of scaleup of CCS technologies using the Storage Readiness Index (Consoli et al., 2015; Consoli, 2018; Havercroft and Consoli, 2018). A high index rating reflects progressive policies or legal support for CCS projects and reliance on fossil fuels. The storage readiness index identifies Canada, the USA, Norway, the UK and Australia as the countries best prepared for advancing CCS deployment (Band A countries). A further five countries (The Netherlands, China, Denmark, Germany and Japan) are considered Band B countries. The basins that correspond to these Band A and Band B countries are also highlighted in Fig. 4.

For all of the scenarios, we identify whether reservoir pressure is limiting maximum injection rates or if injection rates are limited by the number of sites. In scenarios where maximum rates are pressure-limited for a basin, it means that there is interference between the pressure fields produced at adjacent injection sites. As a result, increasing the spacing between injection sites increases the storage resource of the basin. Where this is the case for a given scenario, we identify the injection rate for a basin as limited by pressure. Injectivity and pressure response in reservoirs is very sensitive to permeability. CO2BLOCK calculations derive from mean permeability, reservoir thickness and reservoir area data that ultimately determines whether a reservoir is classified as pressure-limited in this work. Our definition of pressure-limited reservoirs includes low kh reservoirs.

In sites where injection rates are not limited by pressure, the maximum rate (5 Mt $yr^{-1}$ per site) is deployed at all sites. Storage resource is site-limited in reservoirs where pressure plumes from neighbouring injection sites do not meet. Because there is no pressure interference between sites, there is no change in storage resource as site-spacing increases (Fig. 5). We use this feature to identify injection rates for basins as limited by the number of sites. More injection sites would need to be constructed to increase storage resource of site-limited basins.

Simulations employing scenarios 1 – 12 are repeated for both 30 years and 80 years. These periods represent injection until mid-century and injection until the end of the century, timescales frequently used in climate change mitigation plans.

### 2.4. *Demonstrating our workflow using European basins as a worked example*

To ensure the workflow is clear we describe its application to Europe as an example. As described in the basin scheme devised by Evernick (2021), there are 72 European basins. Within these basins, the Wood Mackenzie database records 8739 separate entries for 6673 different hydrocarbon fields. These fields span 51 European countries. These records can include multiple entries for one hydrocarbon field: measurements taken at different localities within the field or at different stratigraphic levels.

After filtering these data for reservoir measurements deeper than 800 m, 7046 data entries remain. These entries are assigned to the European basin in which they are located and are collated with other data that share the same basin. We then calculate the average geological values for each basin. Other statistical parameters such as the number of data entries, standard deviation and interquartile range are also calculated to determine what geological factor results in the greatest uncertainty.

Not all data entries contain a complete set of data. Entries may be missing porosity or permeability values. If we cannot calculate the average geological values needed for CO2BLOCK from combining the Wood Mackenzie, C&C Reservoirs and USGS databases then we consider that basin as having insufficient data (for example, the Vøring basin in Norway). After calculating the average geological values, we have sufficient data to analyse 36 European basins.

We make estimates for each basin across 12 scenarios at both the mid-century and end-century duration. The $CO_2$ storage resource estimate (Gt) and the limiting factor (pressure or site number) is recorded for every estimate. There are no European basins large enough to house the Scenario 12 injection pattern, with the largest number of sites at the greatest site spacing considered. The maximum storage resource is estimated by Scenario M in which the greatest storage resource for each basin is selected and incorporated into Scenario M. For example, the Central and Northern North Sea basins both adopt the Scenario 10

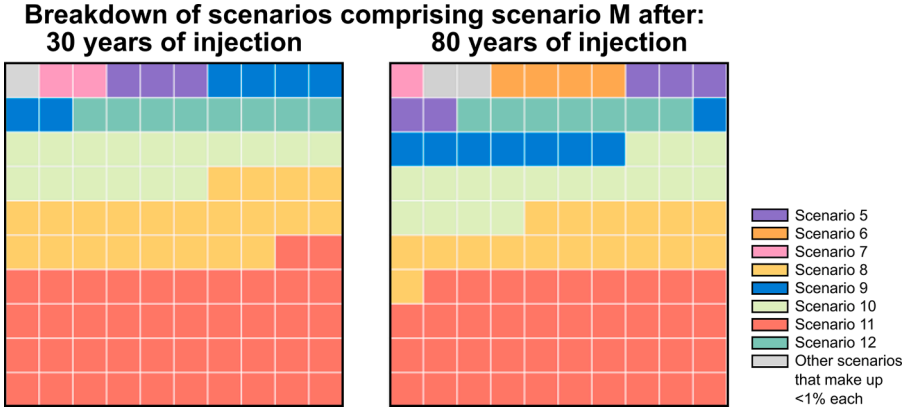



**Fig. 3.** A chart showing the breakdown of which of the original 12 scenarios contribute to the Scenario M storage resource estimates for the next 30 and 80 years respectively. A single square on a chart represents 1 % of basins utilising that scenario, e.g. Scenario 11 generates the greatest storage resource estimate for 42 % of basins when Scenario M operates for 30 years.

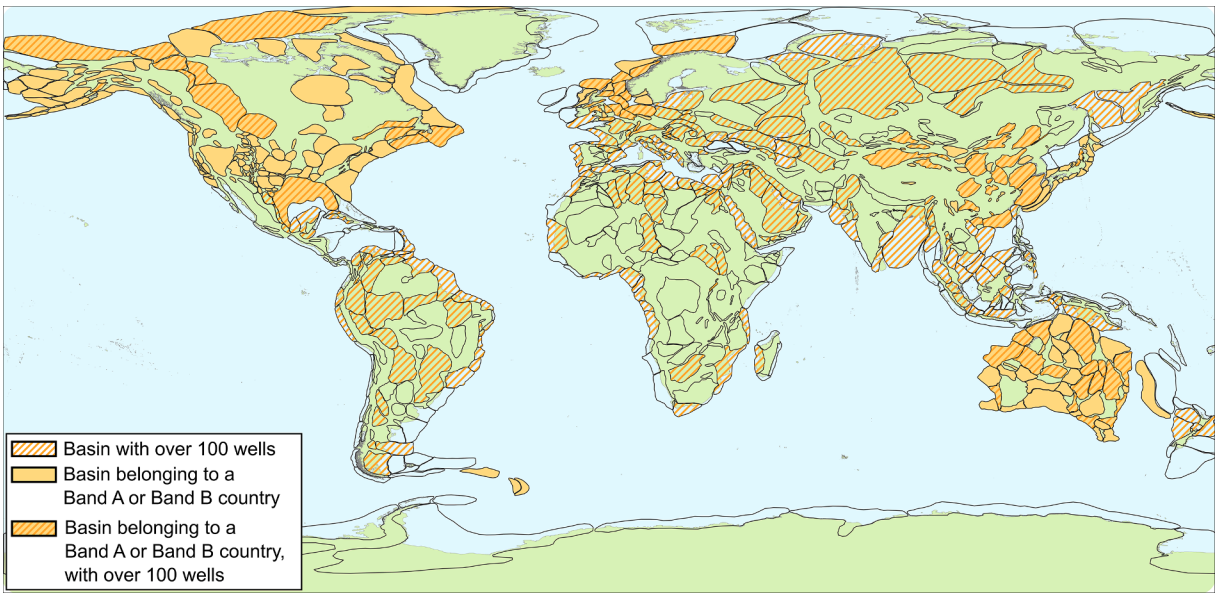



**Fig. 4.** A map showing which basins are included in our analysis where we consider how socioeconomic factors will influence the scope of CCS deployment. See text for details.

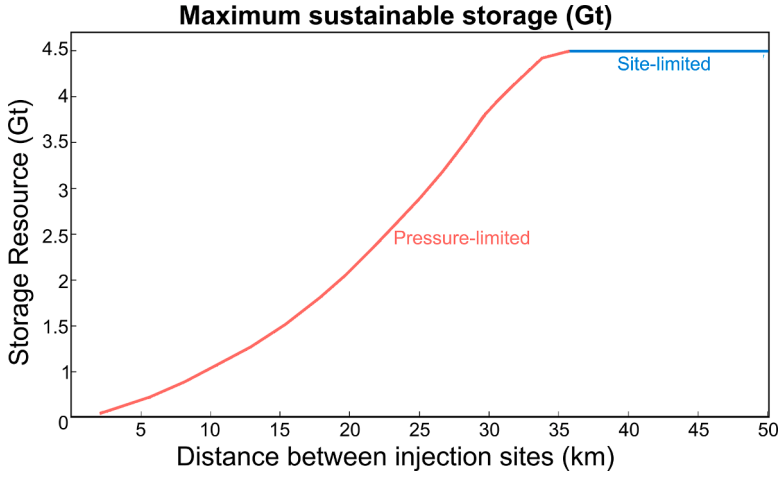



**Fig. 5.** This graph shows the maximum sustainable storage for a given number of injection sites. Pressure-limited and site-number limited basins lie on separate parts of this graph, which allows scenarios of injection site number and spacing in a basin to be assessed. The red line on this graph is an example of pressure-limited storage. Storage resource can be increased simply by increasing the distance between injection sites. The blue line on this graph is site-limited storage, where storage resource cannot be increased without increasing the number of injection sites.

injection pattern, while the Southern North Sea basin adopts the Scenario 8 injection pattern (Table 2).

## 3. Results

Results are presented by region with supporting data and results for each region provided in the Supplementary Material. Our uniform global estimates are those where a single scenario is applied globally. Our greatest storage resource estimate for each basin is amalgamated into Scenario M, which we also present as our maximum storage resource estimate. We analyse these storage resource estimates in relation to projections of CCS deployment from the IEA, GCCSI and IPCC. The IEA report 120 $GtCO_2$ cumulative storage by the mid-century is needed to limit global warming to a 2 °C temperature rise (IEA, 2013, 2020). To meet net zero targets requires 5.6 $GtCO_2$ to be removed and stored annually (GCCSI, 2020). Energy systems models analysed by the IPCC suggest CCS must reach annual storage rates of 10 s $GtCO_2yr^{-1}$(IPCC, 2023). We compare our results against a 10 $GtCO_2yr^{-1}$ threshold to reflect this projection. We also present results where global development varies based on socioeconomic factors such as the presence of a local hydrocarbon industry and the readiness of countries for CCS deployment.

### 3.1. Regional storage resource estimates

The region with the highest storage resource is Asia (Fig. 6). In particular we find Indonesia, China, and India are countries where storage capacities frequently rank in the top ten of our global estimates. In Africa there is high storage resource owing primarily to the Niger Delta, Doba, East Niger, Kwanza, Bove and Senegal-Mauritania basins. In Europe, the countries with the greatest storage resource are the UK and Norway, indicating the importance of the North Sea basin. Storage resource in South America is primarily located in Brazil, within the Amazonas and Campos basins. In the most ambitious scenarios, 9–12

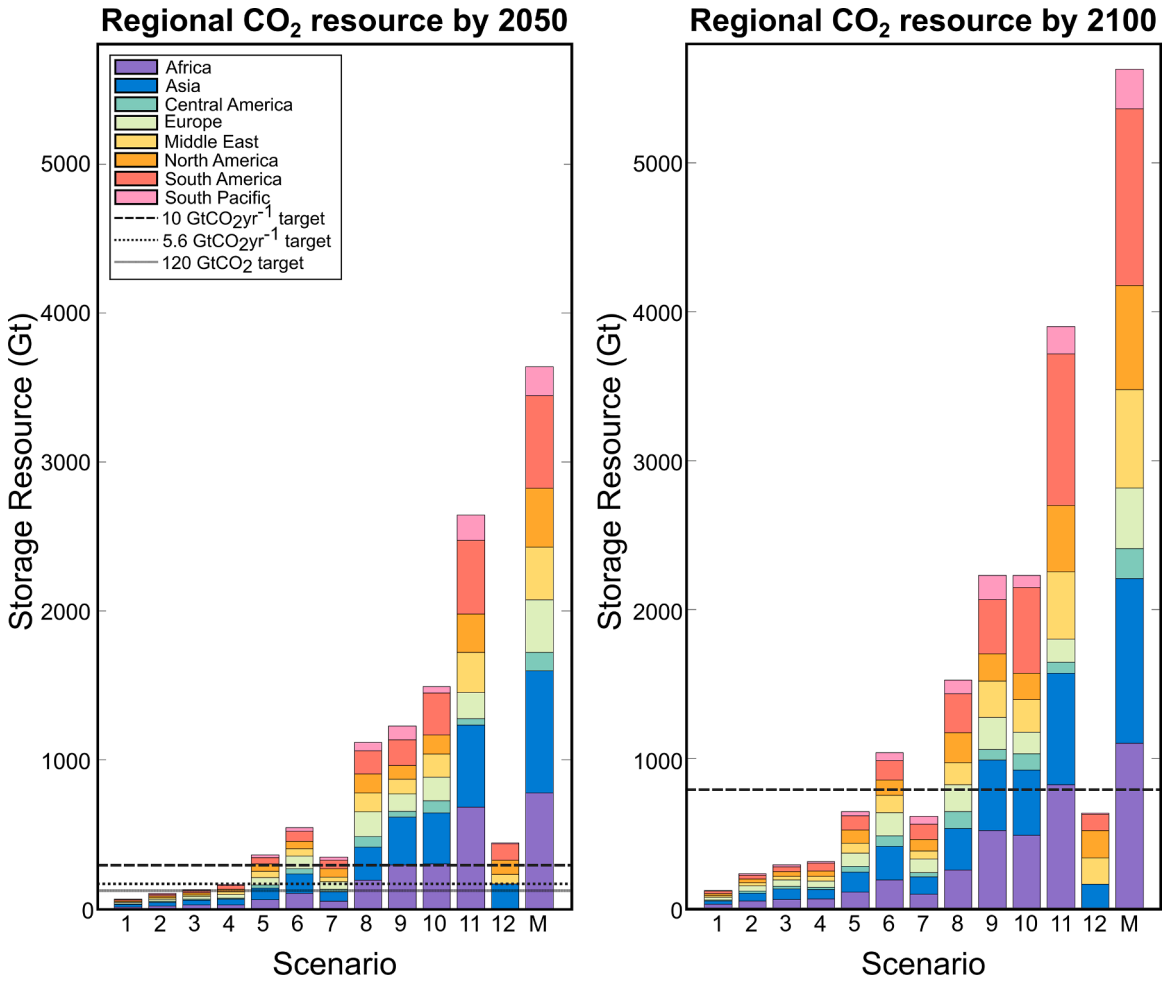



**Fig. 6.** Storage resource estimates for each scenario, colour-coded to highlight the contribution from each region. Horizontal lines show projected deployment from various organisations for comparison (IEA, 2013, 2020; GCCSI, 2020; IPCC, 2023). See text for details.

and the maximum Scenario M (Table 2), the country with the greatest storage resource is Brazil. Central America and the South Pacific have the lowest estimated storage resource. We attribute this to there being fewer basins with data in these regions compared to other continents. Australia generally has a large storage resource, almost always ranking within the top ten countries for storage resource across all scenarios.

Scenario M is achieved with 73,508 injection sites worldwide. It produces our maximum global storage resource estimate of 3640 $GtCO_2$ by mid-century and 5630 $GtCO_2$ by the end of the century.

### 3.2. *Geographic distribution of pressure limitations*

Pressure limits systematically increase as expected with decreased site spacing and with increasing number of sites (Fig. 7). For example, pressure limitations decrease from scenario 1 to 3, from scenario 4 to 6, from scenario 7 to 9 and from scenario 10 to 12. The decrease in these groups represents the impact of keeping the number of sites constant while increasing the spacing between sites. The group of scenarios 4–6 are comparably more pressure-limited than the grouping 1–3 as scenarios 4–6 have more sites than group 1–3.

Pressure limitations are a pervasive feature of large-scale CCS deployment. When injection sites are placed 5 km from each other, pressure always limits the use of the majority of the storage resource. In scenarios 7 and 10 (Table 2), pressure limits 100 % of the global storage resource. Scenario 3, representing the least development of 4 injection sites per basin, spaced 30 km apart, shows the least impact of pressure. However, even in this scenario, almost 30 % of basins are still pressure-limited. This reflects the intrinsic pressure limitations of thinner or less permeable reservoirs.

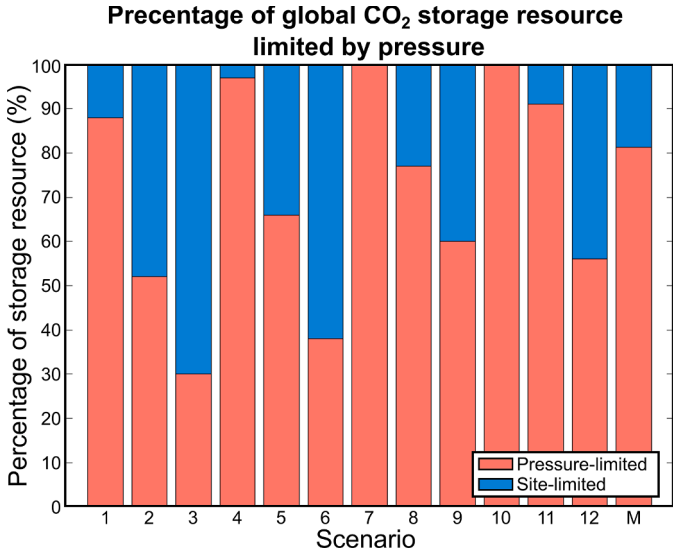



**Fig. 7.** The percentage of global $CO_2$ storage limited by pressure, rather than site number constraints.

The maps in this section and the supplementary information show

the distribution of pressure-limits globally for each scenario. In scenarios with greater site numbers (Scenarios 6, 7, 8, 9 and 11), storage scaleup in North and South America is slightly less limited by pressures than other regions and typically the scenarios contain one or more basins limited by number of injection sites. Within the USA, the Los Angeles and Santa Barbara-Ventura basins are particularly robust to pressure limits, in that they are site limited in most scenarios. For instance, the Los Angeles Basin alone accounts for 96 % of storage not limited by pressures in Scenario 4 (30 years) (Fig. 8a). The dominant factor here is the large reservoir thicknesses in these basins in addition to high reservoir porosity and permeability.

In South America, the Guyana, Amazonas, Campos and Ucayali basins are all particularly robust to pressure limits (Fig. 8b). This is significant as the Amazonas and Campos basins account for a large

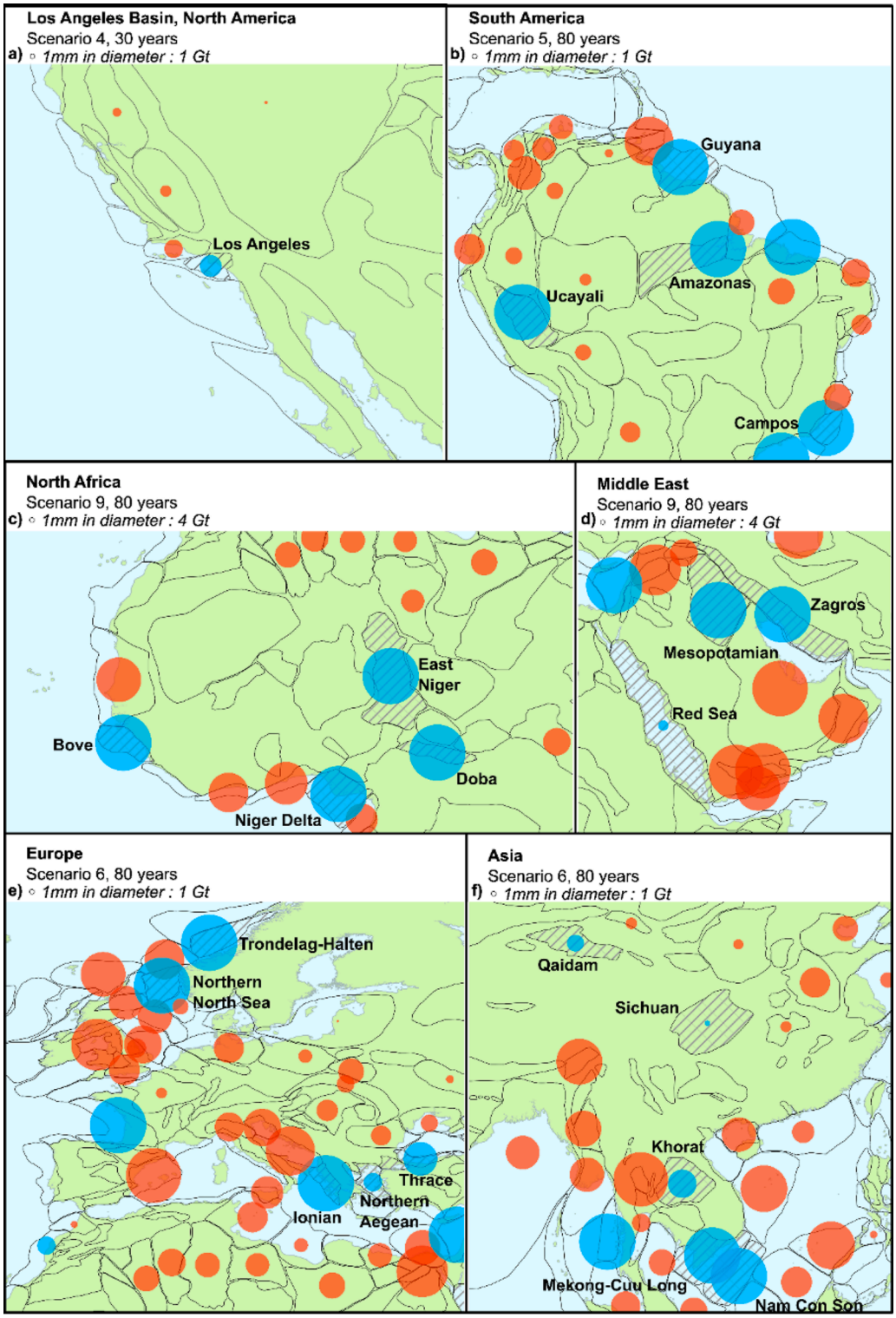



**Fig. 8.** Regional examples of site-limited storage resource. Storage resource estimates are represented by circles, with the diameter scaled to represent the amount of $GtCO_2$ that we estimate could be stored in a basin. Red circles represent pressure-limited storage and blue circles represent storage limited by injection site numbers. Source: Wood Mackenzie.

proportion of the storage resource of this region. In scenarios 4, 7 and 10 the Amazonas and Campos basins alone account for over 30 % of storage resource in South America after 30 years of injection. In Africa, the basins with the largest storage capacities are also the ones that are generally limited by numbers of injection site; the Niger Delta, East Niger, Doba and Bove basins (Fig. 8c). In the Middle East, the Zagros, Mesopotamian and Red Sea basins form a cluster of site-limited basins (Fig. 8d). In Europe, there are two regions that are generally less limited by pressures. These are the Northern North Sea (particularly the Northern North Sea basin and Trondelag-Halten basin) and a cluster of several basins near the Aegean Sea (the Ionian, Northern Aegean, Thrace and West Pontides basins) (Fig. 8e). Asia is generally quite pressure-limited, likely due to the average basin area being smaller. The Vietnamese Mekong-Cuu Long and Nam Con Son basins and the Thai Khorat basin often have significant storage resource limited by the number of injecting sites (Fig. 8f). The onshore Chinese basins, Sichuan, Santanghu and Qaidam basins are also often site-limited. In the South Pacific, the site-limited basins do not cluster together but typically the Browse, Amadeus and Gippsland basins are site-limited. In particular, the Gippsland basin is one of just 12 basins that are site-limited when Scenario 11 continues injecting until 2100.

### 3.3.1. *Socioeconomic barriers to CCS deployment*

Our global storage resource estimates also assume immediate global uptake of CCS technologies. This is a simplification that ignores socio-economic factors that will limit CCS deployment in the 21st century.

Areas with no history of hydrocarbon exploration have more uncertainty surrounding the geological character of the subsurface. This can lead to bottlenecks when developing new CCS projects. To represent this we run a scenario where CCS is only deployed in basins with more than 100 hydrocarbon wells recorded in the Wood Mackenzie wells database. This reduces the number of basins eligible for analysis to 204 (Fig. 4). Restricting deployment to these basins does not significantly impact our storage resource estimates. On average, excluding basins without hydrocarbons experience from our calculations reduces our storage resource estimates by 17 % (Fig. 9). This is because much of the databases we use are derived from the oil and gas industry, so this factor is in effect already accounted for in our data.

Another barrier to CCS deployment is governmental support and the readiness of countries for deployment. To account for this, we run a scenario where CCS is only deployed in countries ranked in the highest two bands by The Global CCS Institute's Storage Readiness Index (Consoli et al., 2015; Consoli, 2015). Band A includes Canada, the USA, Norway, the UK and Australia and Band B includes The Netherlands, China, Denmark, Germany and Japan (Fig. 4). We find the readiness of countries for CCS is the most limiting factor in our storage resource estimates. When we consider Band A and Band B countries, only Scenario 11 and Scenario M meet the 10 $GtCO_2yr^{-1}$ target. In these

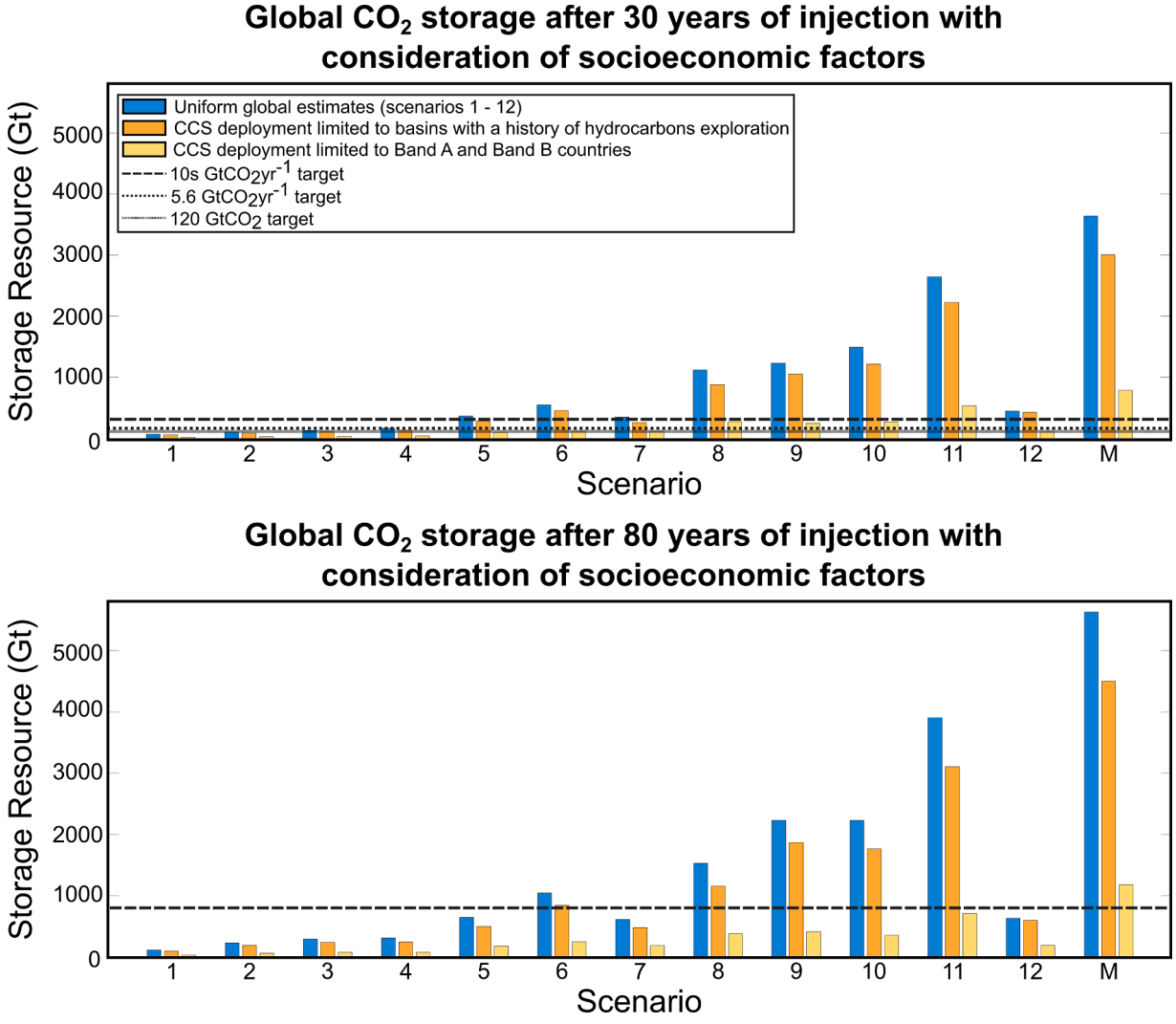



**Fig. 9.** A graph showing how global storage resource estimates decrease when we limit the basins by number of hydrocarbon exploration wells and readiness of basins according to the GCCSI's readiness index.

scenarios 91 % and 81 % of the storage resource is limited by pressures, respectively (Fig. 7).

### 3.4. Comparison with other estimates of storage resource

The storage resource in previously published estimates usually refers to the calculated pore volume available to be saturated by $CO_2$, known as the volumetric approach. Volumetric approaches assume the amount of formation water that can be displaced is unlimited and ignore pressure build up as a limiting factor to injection. This method is also often dependent on an assumed storage efficiency factor. For saline aquifers, storage efficiency factors are generally thought to lie in the range of 0.5 to 4 % (Bachu et al., 2007; National Energy Technology Laboratory and United States Office of Fossil Energy, 2008). This range encompasses nearly an order of magnitude uncertainty, as field data to support storage efficiency factors are scarce and storage efficiency can vary dramatically with geological heterogeneity.

The Oil & Gas Storage Catalogue compiles a wide range of storage resource assessments and produces total storage resource estimates for countries (OGCI, 2022). These compiled assessments may be national surveys focussing on basin potential or more targeted field studies. Where national surveys of storage resource are available, the storage resource estimates reported in the Oil & Gas Storage Catalogue are generally larger than the storage resource estimates for countries calculated using CO2BLOCK in this work.

The volumetric approaches usually produce larger estimates than the pressure-limited approaches because volumetric approaches do not consider the time constraints on storage. For example, the National Carbon Mapping and Infrastructure Plan published by the Australian Government reports 471 $GtCO_2$ storage resource in Australian saline aquifers (Carbon Storage Taskforce, 2009) which is significantly larger than our estimates of 90.2 $GtCO_2$. The National Carbon Mapping and Infrastructure Plan also assumes a storage efficiency factor of 4 %, on the higher end of the range defined by the National Energy Technology Laboratory and United States Office of Fossil Energy (2008) (Carbon Storage Taskforce, 2009). However, development will not be spatially uniform across the storage resource base because of the many other factors controlling where and when a CCS project is developed. These factors include the accessibility of $CO_2$ emissions, suitable geology, permitting regime and project profitability.

Similarly, state-wide volumetric estimates of the saline aquifer storage resource in the USA are greater than estimates produced by the pressure-limited approach. The estimates compiled in the Oil & Gas Storage Catalogue have a combined total of 8613 $GtCO_2$ (NETL, 2015). The majority of these estimates are state-wide volumetric estimates produced using a storage efficiency factor of 2 %. Where data are available local assessments and case studies use other approaches, but these make up a minority of storage resource estimates. The maximum storage resource achieved after 80 years of injection (Scenario M) estimated using CO2BLOCK for the USA is 699 $GtCO_2$. Storage resource estimates are significantly lower when we use CO2BLOCK to consider the role of pressure in limiting injection rates.

Our resource estimates for the USA are closely comparable with storage resource estimates produced by other simple dynamic models evaluating pressure-limited storage. Szulczewski et al. (2014) use a dynamic model to evaluate the storage resource of the Fox Hill Sandstone in the Powder River basin, USA. They find pressures limit CCS deployment for the first 101 years of injection and that the ultimate storage resource available is 3.2 $GtCO_2$. Our maximum storage resource estimate for the Powder River basin is also 3.2 $GtCO_2$. We find this storage can be filled after 80 years of injection by sites in Scenario 8.

Pressure-limited estimates for the UK and Norway are the same order of magnitude as those compiled in the Oil & Gas Storage Catalogue (OGCI, 2022). For the UK, our maximum storage resource estimate after is 71.2 $GtCO_2$, only 6.4 $GtCO_2$ less than the 77.6 $GtCO_2$ estimated in OGCI (2022). CO2BLOCK has previously been used in another study to estimate the storage resource of selected reservoirs offshore from the UK (De Simone et al., 2021). This study uses data from the $CO_2$ stored database (Bentham et al., 2014) that records geological characteristics of potential $CO_2$ storage units. They find 25 sites capable of storing a combined total of 140 $GtCO_2$ in 30 years. Consideration of uncertainty in first order geological parameters such as permeability, leads to an uncertainty of one order of magnitude in the estimates of De Simone et al. (2021). Our maximum mid-century estimate for UK basins of 71.2 $GtCO_2$ lies within this range of uncertainty. Another explanation for the disparity with our results is the use of higher porosity and permeability values in the $CO_2$ stored database. The Wood Mackenzie database from which we derive our porosity and permeability values includes measurements from a larger number of reservoirs, including those not shortlisted in the $CO_2$ stored database. Additionally, De Simone et al. (2021) evaluate saline aquifers located in the UK offshore system, some of which are labelled as belonging to Norway in our study. For example, the Cormorant sites that account for approximately 43 $GtCO_2$ storage after 30 years of injection in the De Simone (2021) study are located in the Northern North Sea basin, which we allocate to Norway because the majority of the basin area is closer to Norway than the UK.

The pressure-limited approach produces an estimate of 80.7 $GtCO_2$ for Norway. This is very similar to the 79.4 $GtCO_2$ reported in the Oil & Gas Storage Catalogue (OGCI, 2022). This estimate is derived from the Norwegian $CO_2$ Storage Atlas, which primarily uses the volumetric approach to estimate storage resource. Storage efficiency factors are assigned using tailored reservoir simulation models or suitable reservoir analogues (NPD, 2014). Estimates of the total pore volume also incorporate geological data and, where possible, are based on seismic and well data (NPD, 2014).

Another discrepancy arises from the lack of published resource surveys for certain regions. For example, we find potential for 1189 $GtCO_2$ storage in Brazilian basins. As there is no published report on the potential for CCS in Brazilian saline aquifers, the Oil & Gas Storage Catalogue cannot record a quantitative estimate of storage resource (OGCI, 2022). This issue also applies to the Middle East and Africa. Only Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates are included in the Oil & Gas Storage Catalogue. However even these reports are not extensive. For example, despite having an extensive hydrocarbons industry suggesting geology suitable for CCS, there is no public evaluation of the storage resource in Oman (OGCI, 2022). Using the pressure-limited approach produces a maximum estimate of 35.6 $GtCO_2$ storage in Oman after 80 years of injection. Similarly, while there are no published national estimates of the Saudi Arabian storage resource, the pressure-limited approach estimates 177 $GtCO_2$. For Sub-Saharan Africa, only Mozambique, Nigeria and South Africa are included in the Oil & Gas Storage Catalogue (OGCI, 2022). Our pressure-limited approach has been applied to estimate storage resource for 21 African countries not included in the Oil & Gas Storage Catalogue.

### 3.5. Implication for well numbers

Injection sites do not directly correlate with well numbers. An injection site could represent a cluster of injecting wells that have a cumulative injection rate of up to 5 $MtCO_2yr^{-1}$. The 12 scenarios presented here imply numbers of injection sites ranging from 992 to over 90,000.

Analysis using historical industry data to estimate storage potential on offshore continental margins suggests that between 10,000 and 14,000 wells are needed by 2050 in order to meet the targets outlined in the Paris Agreement (Ringrose and Meckel, 2019). The wells described in this latter model have an average injection rate of 0.57 Mt/yr, so roughly nine of these comprise one of our injection sites injecting at the maximum rate. This indicates that 1100 – 1600 injection sites would be needed. For reference, the Wood Mackenzie database records 16,809 registered hydrocarbon wells in the North Sea. This equates to 1900 injection sites in our analysis.

Our scenarios 1, 2 and 3 (Table 2) imply 992 injection sites each, just

below the lower range of this projection. If Scenario 3 is applied globally, then 128 $GtCO_2$ can be stored after 30 years. This meets the 120 Gt projection outlined by the IEA (2013) with only 30 % of the global storage resource being limited by reservoir pressure. Meeting the 10 $GtCO_2$ $yr^{-1}$ projection requires a larger scaleup of infrastructure. Scenario 11 requires 50,800 injection sites and Scenario M requires 73,508 injection sites. Although injection sites do not directly correspond to well numbers, if with four injection sites in a fully developed CCS project like Endurance, Scenario M implies 18,377 CCS projects. Alternatively, using the approximation as above where one of our injection sites comprises 9 wells as an upper limit, Scenario M implies up to 661,572 wells. This is an upper estimate as this approximation is for sites injecting at the maximum rate. This number of wells is more than two-thirds of the 920,000 wells recorded in the Wood Mackenzie wells database, which implies substantial CCS infrastructure, approaching that of the current upstream oil and gas industry.

## 4. Discussion

### 4.1. Uncertainties

There are major uncertainties on geological parameters used as inputs to the model. De Simone and Krevor (2021) evaluated the effect that uncertainty from first order geological parameters has upon CO2BLOCK outputs and found the outputs varied by around one order of magnitude. We use the same types of geological data, and our range of uncertainty is similar.

We assign a maximum injection rate be injection site of 5 $MtCO_2$ $yr^{-1}$ in order to provide consistency in the analysis and establish a basis for comparison. 5 $MtCO_2$ $yr^{-1}$ also corresponds to the $CO_2$ capture a typical coal-fired power plant equipped with capture technology is capable of (IEA GHG and ZEP, 2010). This is perhaps a conservative limit. In 2021, Petrobras' Lula CCS project injected $CO_2$ at a rate of 8.7 Mt $yr^{-1}$ (GCCSI, 2022) into the pre-salt oilfield in the Santos Basin, Brazil. Similarly, the Bluebonnet Sequestration Hub in Texas anticipates achieving sequestration rates between 8 $MtCO_2$ $yr^{-1}$ and 26 $MtCO_2$ $yr^{-1}$ (OFECM, 2023). ExxonMobil's proposed CCS Innovation Zone in Texas reports potential storage rates of 50 $MtCO_2$ $yr^{-1}$ using multiple injection wells. However, this project is without a confirmed start date (Hague, 2024).

Our estimates also do not account for the time required to develop projects and scale up CCS technologies. As a result, these results should be considered an upper bound on what may be achievable over a given timescale. Much of the Wood Mackenzie and C&C Reservoirs data derive from hydrocarbon fields, meaning that we are sampling reservoirs that are proven to be suitably porous and permeable, compared to sparsely sampled saline aquifers. This results in us using potentially optimistic values of porosity and permeability. Regions with a history of hydrocarbon exploration and production are already populated by many wells, a proportion of which will have integrity issues. Surveys of documented abandoned wells on the Norwegian shelf found up to 38 % of wells suffered integrity failures (Böttner et al., 2020; Davies et al., 2014). Our storage resource estimates only decrease by 17 % on average when we ignore basins without substantial hydrocarbons industry (Fig. 9). This highlights that most of our storage resource is located in these pre-drilled reservoirs, emphasising the potential risk that future CCS projects face from unsecured legacy wells. In the case of large-scale CCS deployment areas with existing wells will likely be proximal to projects.

There are additional geological uncertainties that are not addressed in this paper. We do not account for any geological features below the basin-scale, such as faulting and the stratigraphic distribution of reservoirs and seals that may give rise to basin compartmentalisation. If faults are sealing and reservoirs are separated by seals, such that basins are divided into smaller compartments, this will increase pressure limitations and isolated structural-stratigraphic compartments will require their own injection sites to develop CCS projects. Ignoring compartmentalisation in this study also indicates that our results will systematically overestimate what is achievable in a given basin.

Ultimately, these uncertainties mean that our estimates can be considered an upper bound. This paper estimates an injectivity-limited storage resource at the basin scale and is not designed to highlight specific areas for storage sites. Individual site selection and development requires specific studies to identify local pressure constraints within basins.

### 4.2. Implication of pressure limitation

If CCS development progresses at a similar rate as over the period 1996–2020, pressure limitations are unlikely to limit deployment. For example, if a project the size of the Endurance project being developed in the Southern North Sea were deployed in all 248 basins analysed in this study, 70 % of the storage resource remains site-limited. However, in any scenario where we successfully meet the 10 $GtCO_2yr^{-1}$ median of IPCC Assessment Report 6 projections and account for the readiness of countries for CCS deployment, at least 81 % of the storage resource is pressure-limited.

This analysis did not consider the mitigating impact of large-scale pressure management systems. Pressure management systems include monitoring systems able to detect changes in wellhead and bottom-hole pressure measurements and strategies for brine extraction. These additional pressure management strategies increase the costs of CCS projects (Harto and Veil, 2011; Davidson et al., 2014). However, they may be important for achieving projected scales of storage.

## 5. Conclusions

We have used the dynamic tool CO2BLOCK to produce storage resource estimates for 248 basins globally. We estimate a maximum global storage resource of 3640 $GtCO_2$ after thirty years of continuous injection, increasing to 5630 $GtCO_2$ at the end of the century. Our approach to calculating storage resource and storage rates is consistent across the globe, allowing for comparison between regions and a methodical calculation process independent of storage efficiency factors.

We constrain injection rates based on pressure build up in reservoirs and identify regions where pressure-limited injection rates may define the scope of CCS deployment. From the perspective of economic development without injectivity constraints there is sufficient storage resource available to meet CCS targets (Zahasky and Krevor, 2020). However, ignoring injectivity limitations has resulted in large storage resource estimates that imply unrealistic capacities can be utilized in mitigating climate change. Our storage resource estimates represent an update to volumetric-based estimates that suggest in excess of 10,000 Gt of storage resource available. Although helpful in ongoing $CO_2$ removal, a large proportion of this reported storage resource is inaccessible within the century. Our storage resource estimates correspond to injection rates of 121 $GtCO_2yr^{-1}$ after thirty years of continuous injection and 70 $GtCO_2yr^{-1}$ at the end of the century. These injection rates are far greater than total deployment envisioned in climate change mitigation roadmaps. Our results emphasize the importance of recognizing the injection rates and timescales needed to access storage resource.

When CCS deployment is limited to basins with a history of hydrocarbon exploitation, our maximum storage resource estimate decreases to 3000 $GtCO_2$ (100 Gt $yr^{-1}$) at the mid-century and 4500 $GtCO_2$ (56 Gt $yr^{-1}$), at the end of the century. This reduction is still far in excess of deployment projections. We identify the greatest barrier to meeting CCS projections as the readiness of countries. When CCS deployment is limited to the countries ranked in the top ten by the Readiness Index, meeting CCS targets is very challenging. In this case, we find our maximum storage resource estimate decreases to 780 $GtCO_2$ (26 $GtCO_2$ $yr^{-1}$) at the mid-century and 1177 $GtCO_2$ (15 $GtCO_2$ $yr^{-1}$) at the end of the century. Although these latter results fall within the range of

projected deployment in the IPCC and IEA, only our two most ambitious scenarios achieve this threshold. Those scenarios (Scenario 11 and Scenario M) employ over 50,000 injection sites each and result in over 81 % of the storage resource being pressure-limited. Even reaching the less ambitious mid-century targets presented by the IEA and GGSCI are challenging once the readiness of countries is considered and our scenarios that reach these lower thresholds require at least 20,000 injection sites (IEA 2013, IEA 2020; GCCSI, 2020). For CCS deployment to meet projections within the current geography of CCS readiness would require injection patterns characterized by closely spaced injection sites that are subject to pressure limitations.

## Supporting information

List of basins

## Disclaimer

The authors would like to clarify that the capacity estimations presented in this work do not reflect the methodologies used for $CO_2$ capacity calculations in Wood Mackenzie's Lens Carbon solution.

## CRediT authorship contribution statement

**Alexandra Smith:** Conceptualization, Methodology, Validation, Formal analysis, Writing – original draft, Writing – review & editing, Software, Investigation, Visualization. **Gary Hampson:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Sam Krevor:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Co-author currently serves as editor-in-chief for the International Journal of Greenhouse Gas Control to which we are submitting - S.K. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijggc.2024.104220.

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